THE FAIRNESS IMPOSSIBILITY

Why "we made it fair" is never a finished sentence

A VADEMECUM



Several precise definitions of fairness, each reasonable, and the mathematics says you cannot satisfy them all at once.

What it is

"Fair" is not one property but several precise, competing ones. Choosing which to satisfy is a value decision, not a technical default, and it cannot be dodged.

DEFINITIONS OF FAIR

- Equal selection rates across groups
- Equal error rates across groups
- Calibration: a score means the same for all
- Similar people treated alike

THE CATCH

- Each is reasonable alone
- They pull in different directions
- When base rates differ, some can't co-exist
- "Fair" forces a choice between them

The tell: "we made it fair" hides which fairness, and which ones it gave up.

Why the conflict is real

This isn't sloppy engineering. It's a mathematical fact, in three moves.



Groups differ in base rates

When a trait appears at different rates across groups, one threshold can't keep every definition equal at once.



The metrics collide

You can't hold equal error rates and calibration together when base rates differ. Improving one worsens the other.



So a choice is forced

No model escapes it. Someone decides which fairness to keep and which to give up, openly or by default.

The heart of it: two "fairs," both right

The COMPAS risk-score fight is the textbook case, and neither side was wrong.



ProPublica's fairness

- Black defendants were more often wrongly flagged high-risk.
- Unequal error rates.
- A real unfairness.



Northpointe's fairness

- Each score meant the same re-offend rate across races.
- Calibration held, a real fairness too.

Both were true at once. With different base rates you cannot have both: the disagreement was a choice, not a bug.

Sources: ProPublica, 'Machine Bias' (2016); Chouldechova (2017); Kleinberg, Mullainathan & Raghavan (2016).

The deeper trap: formal vs substantive

Passing a fairness metric is not the same as being just.



A model can satisfy your chosen definition on paper and still land unjustly, when the data, the proxies, or the labels already encode the inequality you are measuring. Formal fairness can mask substantive discrimination.



So a green metric is where the question starts, not ends. Ask what the label truly measures, and whether treating unequal starting points "equally" is itself fair.

The test: fair how, and who pays?

Don't accept "it's fair." Ask which fairness, chosen by whom, at whose expense.

Which definition?

Which fairness did this model optimise, and which did it trade away?

Who chose?

Was the trade-off decided openly, or buried in a default threshold?

Who bears the error?

Which group carries the false positives or negatives you tolerated?

Formal or real?

Does the metric hold because the outcome is just, or only because the maths balances?

Every question is a "who decides, who pays" question. Fairness isn't a box you check; it's a trade-off you own.

Vademecum — the essentials

Definition

"Fair" is several precise, competing properties, not one default.

The fact

When base rates differ, some fairness definitions cannot co-exist. Proven, not opinion.

The case

COMPAS: ProPublica and Northpointe were both right, measuring different fairnesses.

The caution

Formal fairness can mask substantive discrimination: a passed metric can still be unjust.

The real skill

Not achieving "fairness," but naming which one you chose and who carries the cost.

The test

Fair how, chosen by whom, at whose expense: formal or real?

"We made it fair" is a choice someone made, and should have to defend.